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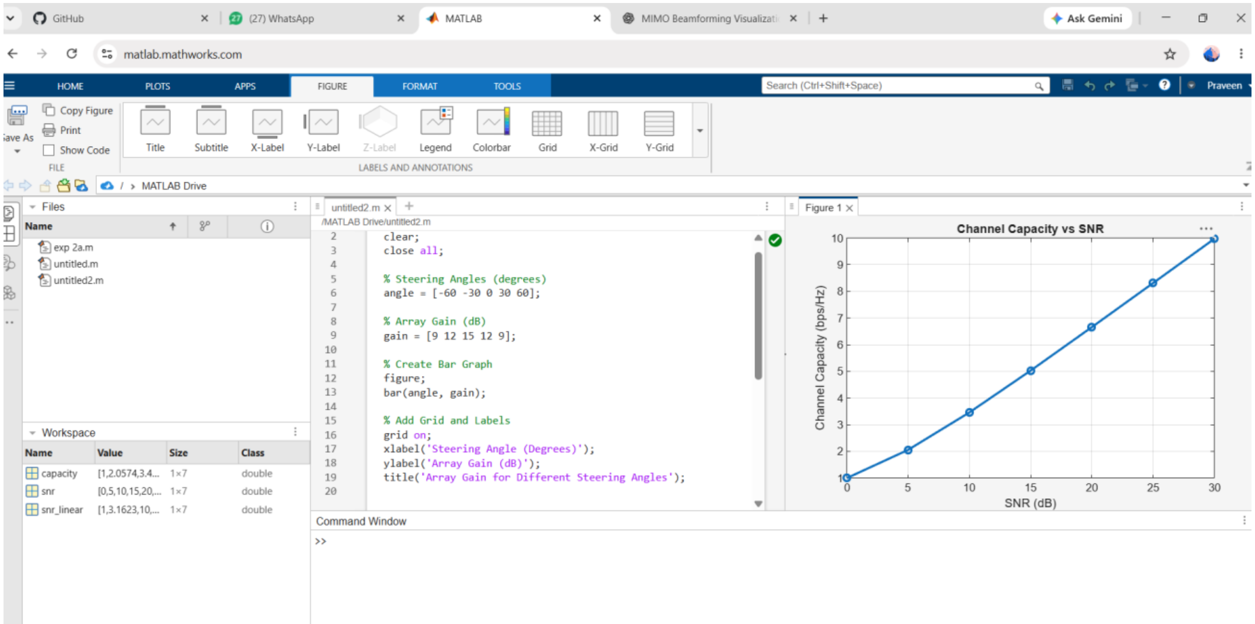
EXP 9

```
clc;
clear;
close all;

% SNR values in dB
snr = 0:5:30;

% Convert SNR from dB to linear scale
snr_linear = 10.^(snr/10);

% Calculate Channel Capacity (bps/Hz)
capacity = log2(1 + snr_linear);

% Plot Channel Capacity vs SNR
figure;
plot(snr, capacity, '-o', 'LineWidth', 2, 'MarkerSize', 6);

grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```



OBJ 2

clc;

clear;

close all;

% SNR values (dB)

snr = 0:5:30;

% Convert SNR from dB to linear scale

snr_linear = 10.^(snr/10);

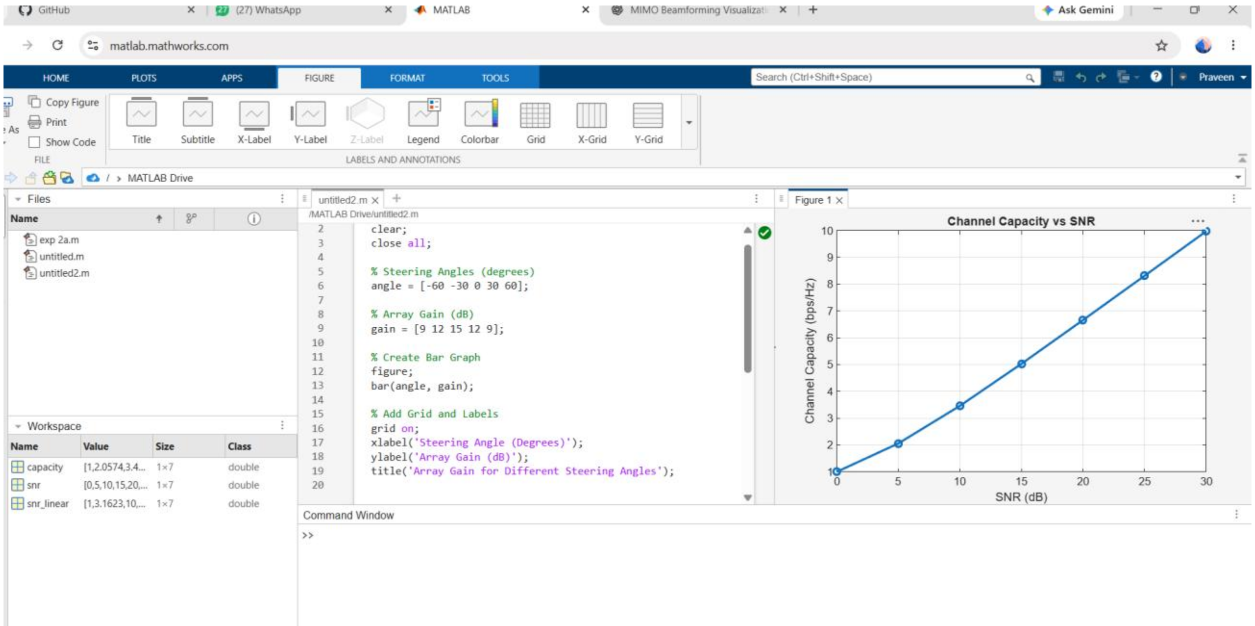
% Calculate Channel Capacity (bps/Hz)

capacity = log2(1 + snr_linear);

% Plot the graph

figure;

```
plot(snr, capacity, '-o', 'LineWidth', 2, 'MarkerSize', 6);


grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```



OBJ 3

```
clc;
clear;
close all;


% Number of Antenna Elements
antennas = [4 8 16 32 64];


% Antenna Gain (dB)
gain = 10 * log10(antennas);
```

```
% Plot Bar Graph
figure;
bar(antennas, gain);

grid on;
xlabel('Number of Antenna Elements');
ylabel('Antenna Gain (dB)');
title('Antenna Gain for Different Antenna Array Sizes');
```